Subsidiarity for Comms

Disconnect from central, connect local, starting with comms · from Oct 24–25, 2026 memos · Edge Collective

The idea: the answer to a harder internet is the same as the answer to climate. **Disconnect from centralized services** wherever you can, and **build local ties**, material and social. Work through the basics (food, water, shelter, heating and cooling, comms, energy) at **household, block and town** scales, over **3 days → 3 years**. Comms is the first domain to tackle. The old local-first / peer-to-peer ideas are now **urgent**, and the AI hacking arms race points further, to **local *and analog* first**. In practice: handle each need at the most local level that can manage it, and only go out to the open internet through a careful **airlock**. A small group with coding superpowers can **prototype this fast** and test it at **offline-first gatherings**.

THE BIGGER FRAME

Disconnect from central, connect local

Same answer as climate

Before any of this, the conclusion on climate was already to lean less on centralized services and more on community. The internet crisis just **speeds up the same project**, and a lot of that project is fun and joyful.

Off-grid is normal

It was the default for **most humans for most of history**, and still is for a lot of people today. It only feels extreme because we sit at the far end of one particular way of living.

Think camping, not catastrophe

Put in modest terms and compared to camping, it isn't scary. A **small input early** gets you a lot of leverage against short disruptions.

Buy a learning curve

If the grid is down for a week and you have three weeks ready, you're **calm**. You can spend that time talking, gathering information and planning for a longer outage, instead of scrambling for basics.

One of the better timelines

Today's attacks are **loud and detectable**. The worse version is quiet, sneaky attacks, and some sleepers may already exist. Loud attacks bring **widespread awareness** of the risk while there's still time to respond.

New arrivals

Lots of people from very plugged-in lives are interested now, though they never expected to need this. Expect them to **sound freaked out**, and meet them with calm, practical steps.

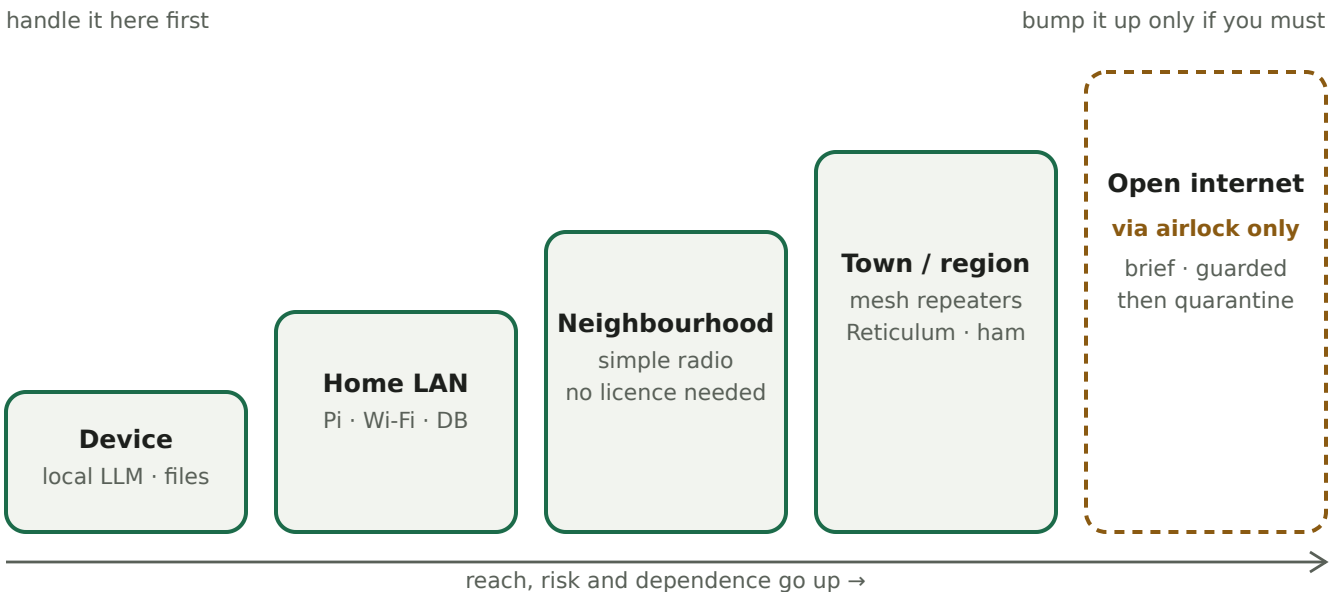
The map: each basic need is a column of cells. Move outward in space and forward in time, one cell at a time.

Unplugged for… →	3 days	3 weeks	3 months	3 years
You + family + friends	camping kit	stores; buys the learning curve	garden, water source, heat	can't do it alone
Block / building	know your neighbours	simple radio net, pooled tools	shared power, water and comms nodes	skills spread around
Town	n/a	mesh across town	food and trade links	real local capacity
Bioregion?	*possibly a stretch, for later*			

Cell contents are rough examples. "The kit buys two to four weeks": the same point as buying a learning curve.

COMMS, FIRST

Subsidiarity for comms



ANALYSIS

Why comms, and why now

Comms comes first

It gives **situational awareness** (what's going on, fast) and **coordination at a distance**. It's also one of the more technical domains, so it's where a little guidance helps most.

Then: a niche worry

Local-first, p2p and data-provenance projects ran on **democratizing power** and avoiding central control. Resilience was a side benefit. Most people saw it as theoretical, with no urgency.

Now: a pressing one

With AI-fuelled attacks on infrastructure, "the internet is over" is all over the feeds. The answer Graeber and others gave about society fits here too: *it wasn't that great to begin with, so build a better one.*

Subsidiarity as the design rule. Deal with each issue at the most local level, closest to where it's felt, and bump it up a level only when that level can't cope. Applied to systems, that means keeping resources on the local network wherever you can, and treating the open internet as a last resort you visit briefly and carefully.

> *"Build a better system in the shell of the old."*

THE ALTERNATIVE

Through vs. around

The main rival view is **"the only way out is through"** (Eric S. Raymond, on the "Bugpocalypse"). The world's code is finite, so its defects are finite, and AI can now red-team and fix them all. It's "given enough eyeballs, all bugs are shallow," with AI as the eyeballs.

Why "through" is a risky bet

The corpus **keeps growing**, because AI writes new code faster than ever. A patch written **isn't a patch deployed**: billions of routers, gadgets and old phones never update. Defenders must win **every time**, while an attacker needs one miss. Every fix is new code. So the scope and timeline are unknowable.

What "around" does instead

Subsidiarity **shrinks what you depend on** to something you can actually audit, and **contains the damage** when something breaks, like firebreaks or quarantine. It works even if the race is lost, or never ends.

They combine

Use the same AI lockpicks on **your own small stack**. "Everyone red-teams their own code" is realistic when your code is a Pi and a mesh, not the whole internet. In PACE terms: keep fixing the big shared system as the primary path, but **never let it be the only one**.

> Fixing everything is a bet you have to win. Subsidiarity is a bet you don't.

THE ESCALATION LOGIC

Local first, and now analog first

1. **Today's models are already enough.** AI is supercharging hacking now, and much of it probably isn't even using the frontier models. People are using the latest public releases to find flaws in software.
2. **It gets worse from here.** The labs are holding back their most capable models, but others (China, open-weight projects) will release theirs.
3. **Defence escalates too.** Better attacks push people to build more capable models to defend, and around it goes to a **fever pitch**.
4. **The systems are too tangled to clean.** They have huge numbers of moving parts that no human can fully reason about, and the **connections between systems** are as risky as the systems themselves. Having AIs find every vulnerability across all of it looks fragile and unlikely to work.
5. **So do what you'd do with a buggy inherited codebase:** start again **small** and build up step by step. It's the same motive behind rewrites in Rust and other designs that are secure from the start.

Local first, redoubled

Every reason we had before for building local-first still holds, and the escalation adds a pressing new one.

Analog first

The same logic points further: **local and analog first**. Paper, voice radio, mechanical tools and face-to-face arrangements have no software to exploit at all. Digital goes on top of an analog floor, not instead of it.

Better anyway

Analog-first is likely better in a lot of ways, including **for our nervous systems**. Less screen time, more attention, more time with people. Security and wellbeing point the same way.

GENERAL IDEAS

The building blocks

Go digital

Basic voice radio covers a neighbourhood. Past that, you need ham or mesh. The real gain is **digital**: send a whole document instead of reading out lists of names. Ham emergency teams already do this with email over radio.

Carry real data

Keep and share **large amounts of data** when it's useful: databases, files, local LLMs. Make it work so that **lots of people** can run it, not just specialists.

The airlock node

One machine goes online, **cut off from the home network**, only for as long as it takes to fetch a resource. It gets sanitized afterwards. Data comes back in by USB stick. Think of it like an Ebola-ward infection-control protocol.

A local LLM on guard

A trusted local model checks the security of the connecting system. It might be a simple one, or the most capable model you can afford to run at home.

Old stuff, with tweaks

No new protocol needed (yet). The work is **R&D on combinations** of existing tools, and the approach stays agnostic about any one technology.

Humans bring taste

The models handle the technical work. People supply the **interest, taste, curation and requirements**. We know the use cases, so we can steer.

Why small stacks

AI makes finding old bugs cheap, so **attack surface** is now the thing to design around. Use **less code and fewer dependencies** (compact Reticulum, not a deep npm tree). Use **dumb file formats**: flatten incoming files before a trusted machine opens them. Remember that radio firmware can be reached by anyone in range, so stress-test it. Keep the whole stack small enough to audit end to end.

THE PROJECT

Edge Collective: offline-first R&D

Kind of project: research and prototyping plus public gatherings, under **Edge Collective** (a better fit than WFS, or we give it its own name later). We build it for ourselves first, then for the people and organizations we know.

Stack to try first

Reticulum (it already has strong demos) → over **mesh / LoRa radio** → with **Hyperdrive** (the Dat project's descendant) on top → plus a **local LLM**. Hyperdrive may be too bloated for secure off-grid links, but a leaner version would now be easy to prototype.

The gathering format

A challenge with **no existing internet infrastructure allowed**: get files from one side of a city to the other. Teams bring Raspberry Pis with local Wi-Fi, and place mesh repeaters on rooftops using a map and walking the route. The same exercise works on a farm or in a suburb.

Places & people

The farm (it's keen, and can host workshops), Dorn's farm, contacts in Burlington and Montpelier, and others. That gives both rural and urban test beds.

Who might fund it

Nonprofits serving **vulnerable populations** who are worried about their safety and security, and now also **high-profile, wealthy organizations and individuals** facing the same exposure. We have relevant backgrounds and past projects to point to.

WHAT'S EXCITING

Why this is the moment

- **We already did the thinking.** Years of local-first work back in the day, and now the urgency has caught up with it.
- **"Somebody should" has become "we can."** Coding superpowers mean a small group can build real infrastructure quickly.
- **Fast, tangible demos.** Reticulum can run a live demo for a group very soon, then lead into brainstorming what to build next.
- **It's fun and it's timely.** A city-crossing, no-internet file relay is a game, a workshop and a field test all at once. "So hip right now."
- **People want it.** Funders and partners across the range, from vulnerable communities to the very wealthy, share the same worry.

NOTES TO CHECK, NOT SETTLED

- **Reticulum on mesh radio is already a thing.** Its RNode firmware runs Reticulum straight over LoRa boards, and NomadNet / Sideband (LXMF) already do store-and-forward messaging and pages. That's a ready-made first demo, and it is an easy first step. MeshCore is a separate stack, not Reticulum.
- **Hyperdrive over LoRa:** the catch is bandwidth. LoRa gives roughly hundreds of bytes to a few kbps, and Hypercore is chatty. A likely pattern: Hyperdrive on fast local Wi-Fi islands, with Reticulum carrying small signed updates or diffs between the islands.
- **Airlock = a "sneakernet" protocol.** Write it as a one-page SOP for gatherings: a wiped live-USB node, read-only transfer media, checksums and signatures on everything that comes back in.
- **Connects to** PACE planning and the power-and-comms Friday meetup (Oct 23) and to an adopt-a-node approach for siting repeaters.

NEXT STEPS

Toward a first gathering

1. **Use cases first:** list 3–5 setups people would actually want (household, farm, neighbourhood, an org with vulnerable members).
2. **Bench demo:** two Pis talking over Reticulum on Wi-Fi, then over RNode LoRa, and send a file across.
3. **Spike:** test Hyperdrive across a Reticulum link, or sketch the leaner alternative. Try a local LLM as the "security guard" on the airlock node.
4. **Draft the gathering:** the no-internet challenge, kit list and repeater-siting map exercise. First run on the farm, then Burlington or Montpelier.
5. **Framing:** a short Edge Collective one-pager covering past projects, the subsidiarity framing and the asks for funders and partners. Decide on a name.

Drawn from dictated memos by Don Blair, Oct 24–25, 2026. Working notes, not a finished essay.